Steel Industry Energy Consumption — Advanced EDA & Modeling

Name	javeria Irfan
Roll No.	AIMLB01-94888
Document Type	Project Documentation

1. Project Overview

This document summarizes the notebook "steel_energy_full_pipeline.ipynb", which analyzes Steel Industry Energy Consumption data and builds a machine learning pipeline to classify the electrical Load Type (Light, Medium, or Maximum) using consumption and power-quality features.

The notebook is organized into two parts: an in-depth Exploratory Data Analysis (EDA) and an advanced multi-model classification pipeline with class-imbalance handling and explainability.

---

2. Dataset Summary

- Source file: Week 2 (DataSet).xlsx
- Granularity: readings recorded every 15 minutes over one year
- Target variable: Load_Type (Light_Load / Medium_Load / Maximum_Load)
- Key features: Usage_kWh, Reactive Power, CO2 emissions, Power Factor, NSM, WeekStatus, Day_of_week

---

3. Part 1 — Exploratory Data Analysis

3.1 Data Preparation

The date column is parsed into a proper datetime object, and cyclical time features (hour_sin, hour_cos, month_sin, month_cos) are engineered so the model can correctly understand daily and seasonal patterns.

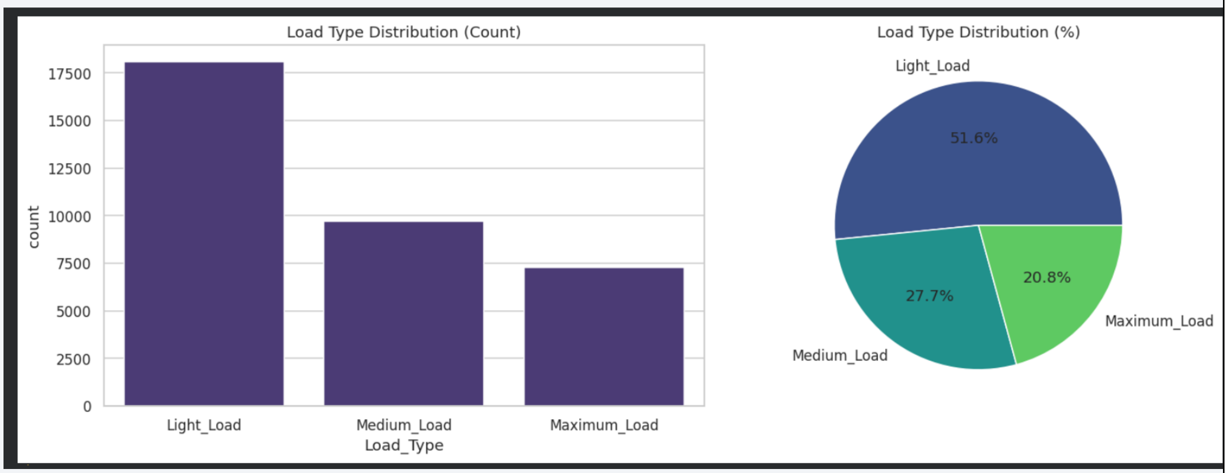
3.2 Data Quality Check

A structured summary table reports missing values, duplicate rows, and data types for every column.

	dtype	missing_count	missing_%	unique_values
date	datetime64[ns]	0	0.0	35040
Usage_kWh	float64	0	0.0	3343
Lagging_Current_Reactive.Power_kVarh	float64	0	0.0	1954
Leading_Current_Reactive_Power_kVarh	float64	0	0.0	768
CO2(tCO2)	float64	0	0.0	8
Lagging_Current_Power_Factor	float64	0	0.0	5079
Leading_Current_Power_Factor	float64	0	0.0	3366
NSM	int64	0	0.0	96
WeekStatus	object	0	0.0	2
Day_of_week	object	0	0.0	7
Load_Type	object	0	0.0	3

3.3 Target Distribution & Class Imbalance

The distribution of Load_Type is visualized, and the imbalance ratio between the majority and minority classes is calculated to justify using SMOTE later in modeling.



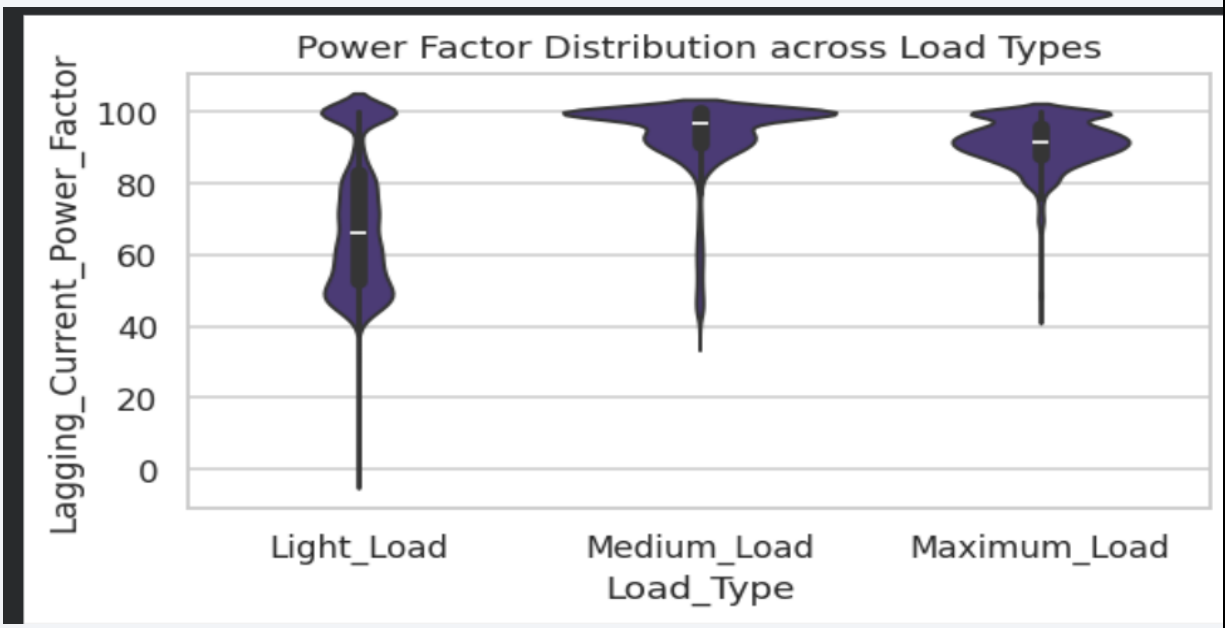
3.4 Outlier Detection

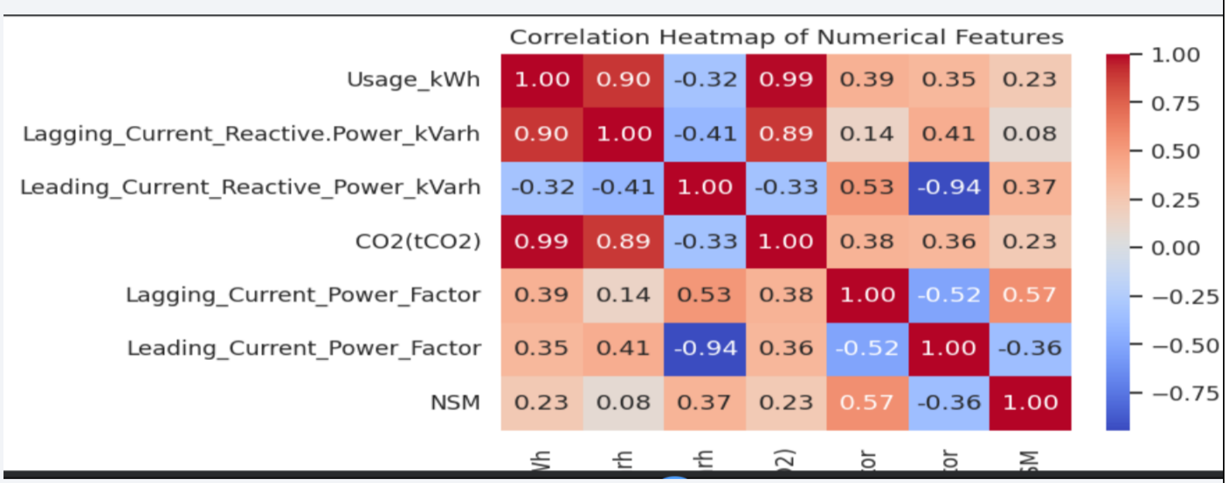
Statistical outliers are identified for every numeric feature using the IQR method and ranked in a summary table.

	outlier_count
Leading_Current_Power_Factor	8327
Leading_Current_Reactive_Power_kVarh	7759
Lagging_Current_Reactive.Power_kVarh	1059
CO2(tCO2)	437
Usage_kWh	328
Lagging_Current_Power_Factor	1
NSM	0

3.5 Univariate & Multivariate Analysis

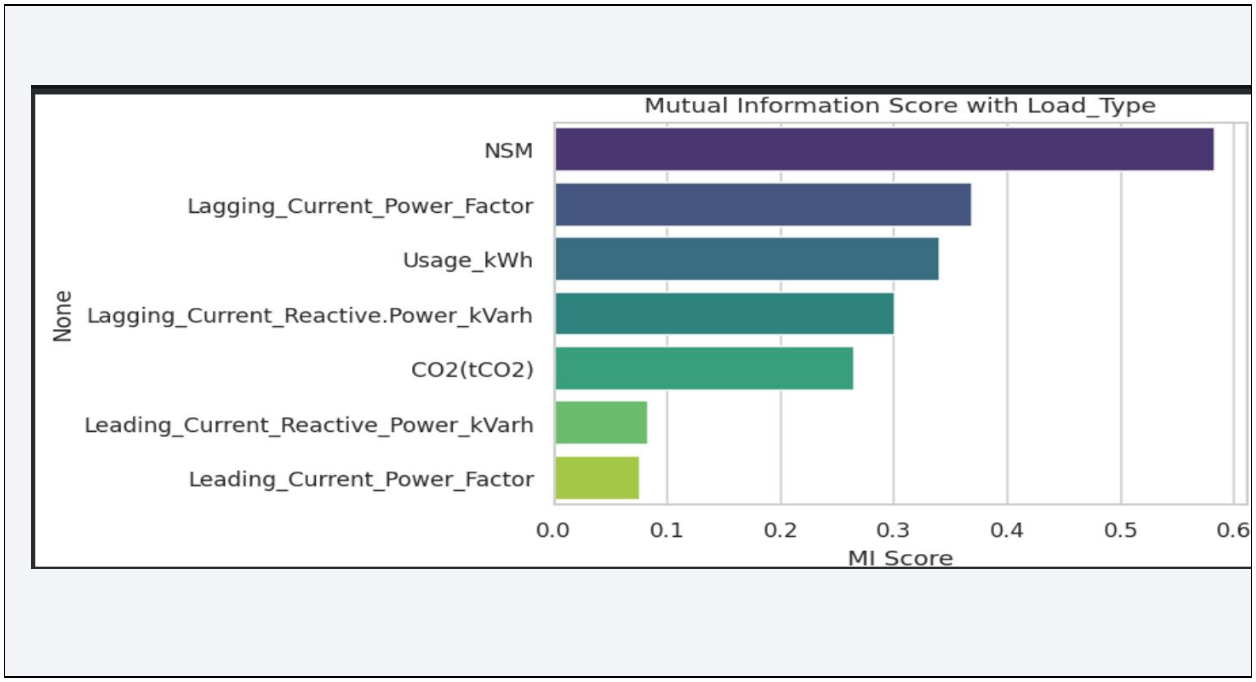
Distribution plots, a violin plot of Power Factor vs Load Type, and a correlation heatmap are used to understand feature behavior and relationships.





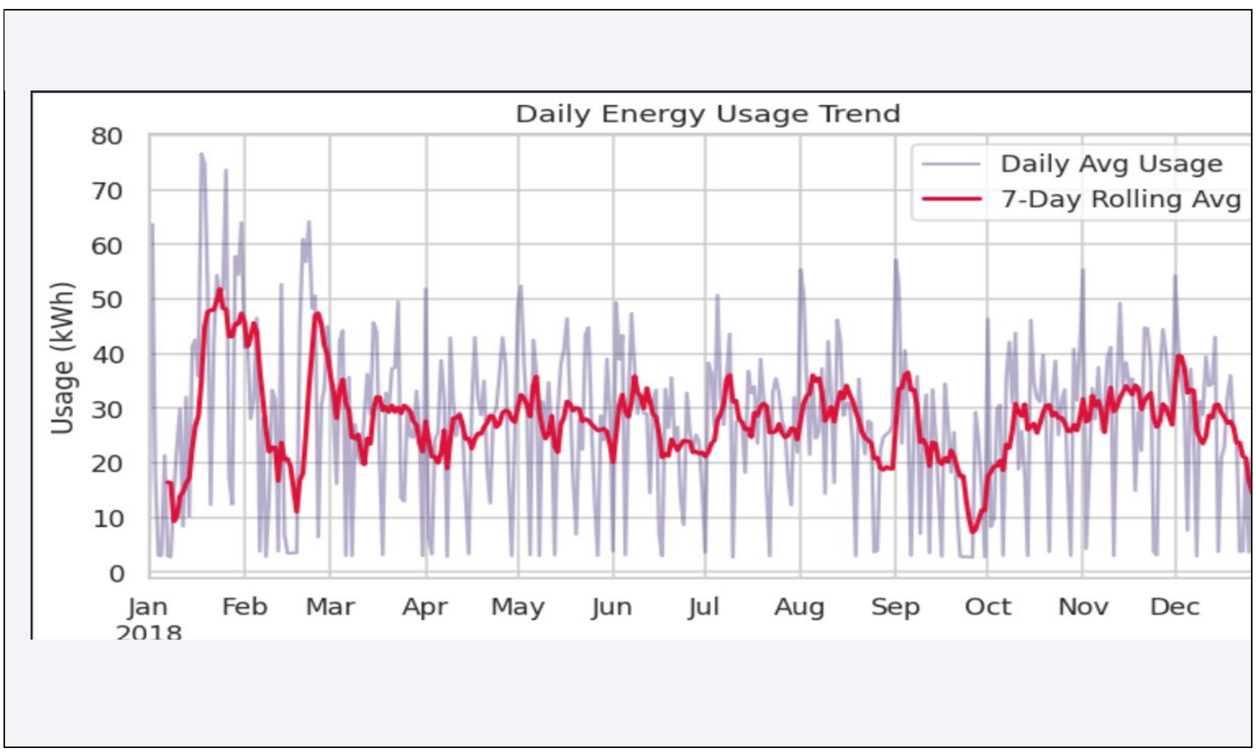
3.6 Mutual Information

Mutual Information scores are computed to capture non-linear relationships between features and the target, going beyond what correlation alone can show.



3.7 Time-Series Trend

A 7-day rolling average of daily energy usage is plotted to reveal the underlying consumption trend across the year.



4. Part 2 — Advanced Modeling Pipeline

4.1 Preprocessing Pipeline

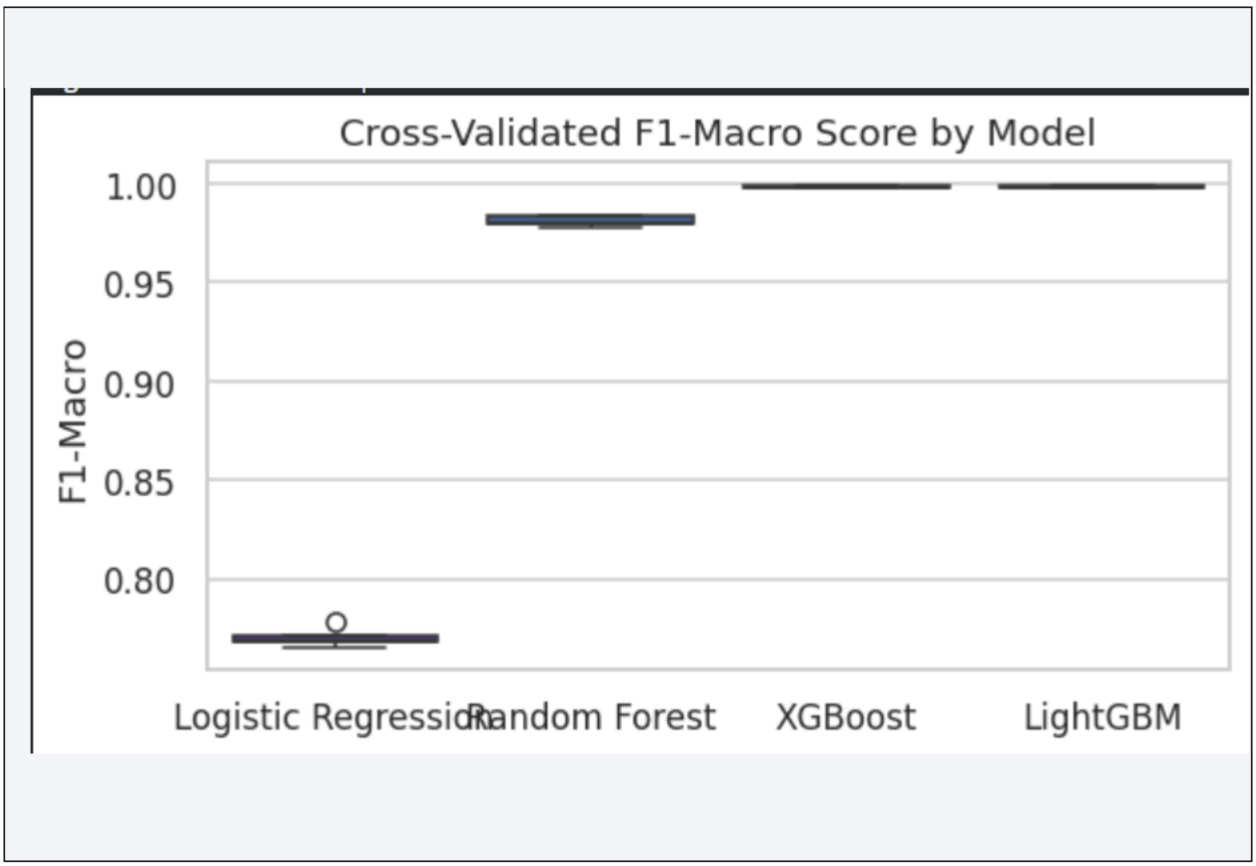
Features are split, encoded (One-Hot for categorical columns, Label Encoding for the target), and passed through a Pipeline combining StandardScaler and SMOTE — ensuring scaling and resampling are applied only on the training data, preventing data leakage.

4.2 Models Compared

- Logistic Regression — linear baseline
- Random Forest — ensemble tree-based model
- XGBoost — gradient boosting model
- LightGBM — fast gradient boosting model

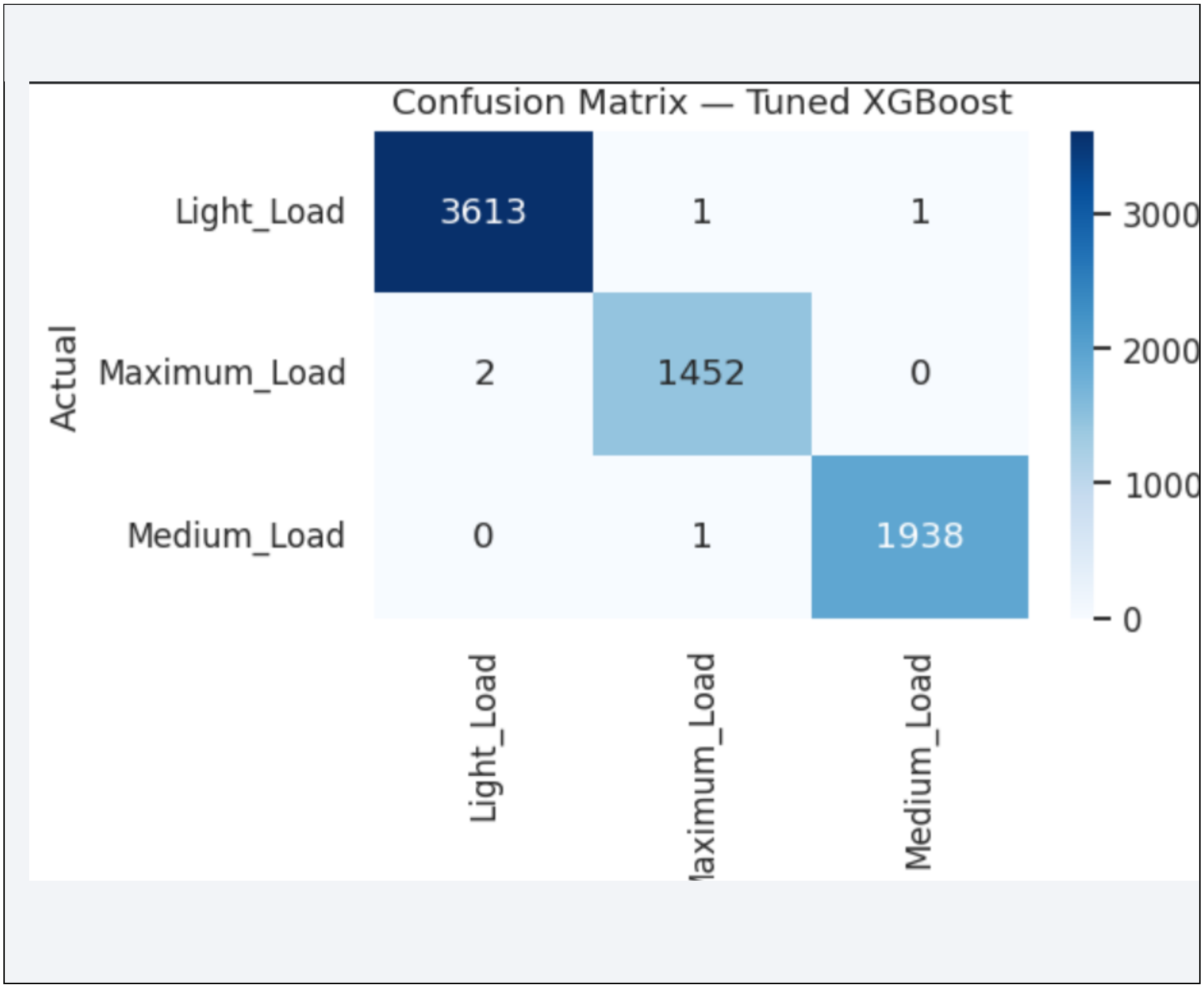
4.3 Cross-Validation & Tuning

Models are compared using 5-fold Stratified Cross-Validation with F1-macro score, which is more reliable than accuracy for imbalanced multi-class problems. The best-performing model (XGBoost) is then tuned using RandomizedSearchCV.



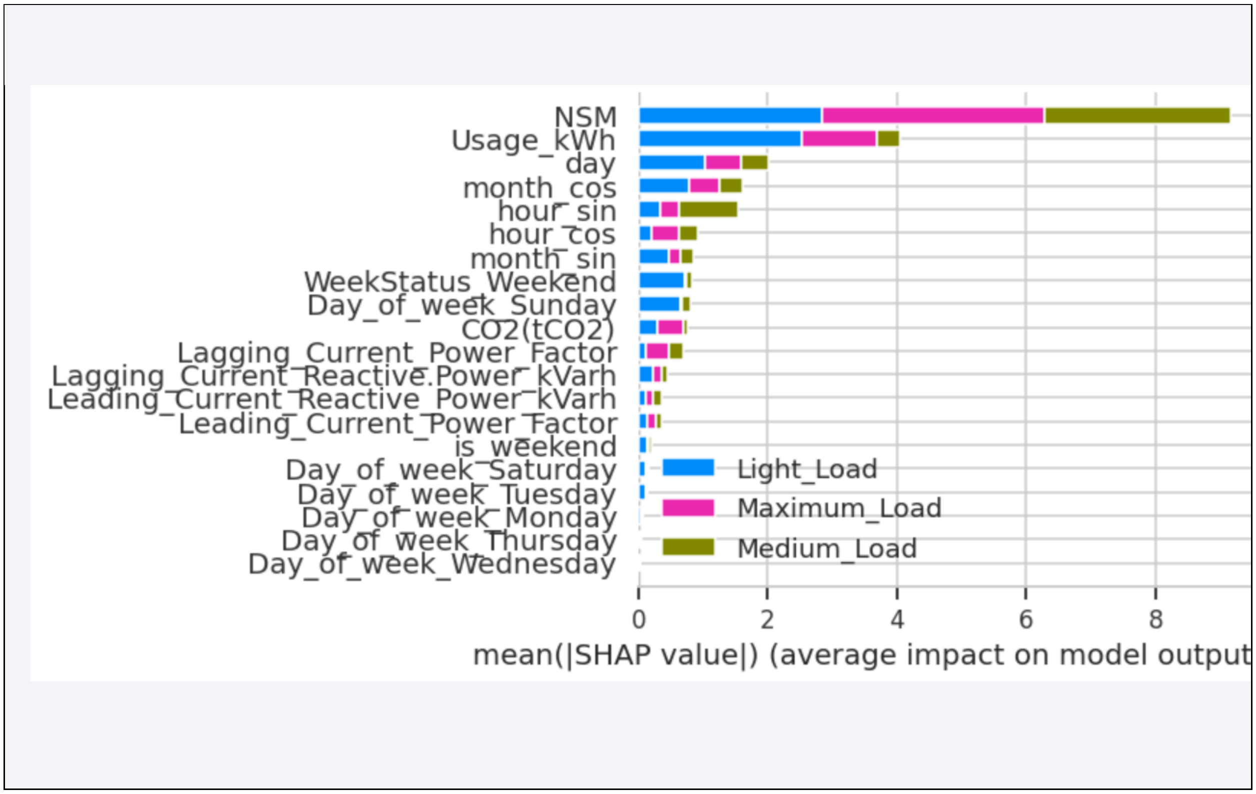
4.4 Final Evaluation

The tuned model is evaluated on a held-out test set using a classification report, confusion matrix, and macro ROC-AUC score.



4.5 Explainability (SHAP)

SHAP values are used to explain individual feature contributions to the model's predictions, identifying Reactive Power and Power Factor as the most influential features.



5. Conclusion

The tuned XGBoost model achieved the best F1-macro and ROC-AUC scores among all tested models. SMOTE improved recall on the minority Maximum_Load class, and SHAP analysis confirmed that Reactive Power and Power Factor are the strongest predictors of Load Type — consistent with electrical engineering domain knowledge.

5.1 Possible Next Steps

- Bayesian hyperparameter optimization using Optuna
- Time-series modeling (e.g., LSTM) given the sequential nature of the data
- Deployment of the tuned model as a REST API using FastAPI